

Traffic Count using YOLOv12 and Faster R-CNN

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Abstract. This report presents a traffic counting system leveraging deep learning models: YOLOv12 and Faster R-CNN. Both models are applied to detect vehicles, followed by tracking and speed estimation. The system is aimed at improving automated traffic surveillance using pre-trained object detection networks.

1 Introduction

Monitoring vehicular traffic plays a key role in smart city planning. Automated solutions based on deep learning have shown high accuracy in object detection tasks. This study explores the usage of YOLOv12 and Faster R-CNN for vehicle detection, counting, and speed estimation.

2 Methodology

2.1 YOLOv12-Based Detection

YOLO (You Only Look Once) is a fast and accurate object detector. We used YOLOv12 for its improved speed-performance trade-off. The following steps were involved:

- Install necessary libraries
- Define detection classes
- Capture video frames and perform detection
- Track objects using ID assignment
- Calculate speed based on frame-to-frame displacement

2.2 Faster R-CNN-Based Detection

Faster R-CNN offers high accuracy in object detection but with slower inference time. It involves:

- Loading a pre-trained ResNet-based Faster R-CNN model
- Using COCO classes to identify vehicles
- Applying detection on video frames
- Estimating speed similar to YOLO-based system

3 Experiments

The models were tested on traffic videos. Both frameworks performed vehicle detection in real-time. Object IDs were tracked over frames to estimate speed using the formula:

$$\text{Speed} = \frac{\text{Pixel Distance} \times \text{Real-World Scale}}{\text{Time Interval}}$$

4 Results

YOLOv12 achieved faster inference with moderate accuracy. Faster R-CNN was more accurate in cluttered scenes but slower. The system successfully counted vehicles and estimated speeds in multiple video tests.

5 Conclusion

This work demonstrates that combining detection and tracking algorithms enables real-time traffic monitoring. Future work can focus on using hybrid models or transformers to improve detection under occlusion.

References

- Redmon, J. et al., "You Only Look Once: Unified, Real-Time Object Detection," CVPR, 2016.
- Ren, S. et al., "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," NIPS, 2015.
- COCO Dataset: <https://cocodataset.org/>